

Northwind Freight - Operations Handbook

Internal reference for operations, support, and engineering staff. Version 3.2, fictional company, for training use only.

1. About This Handbook

This handbook describes how Northwind Freight runs its day-to-day operations. It covers customer refunds, shipping tiers, warehouse safety, data retention, on-call duties, and incident response. Every new hire reads it during their first week, and team leads are expected to know the sections that apply to their area without looking them up.

The handbook is owned by the Operations Excellence team. Changes are proposed through a pull request against the policy repository and must be approved by two section owners before they take effect. A change that alters a customer-facing commitment, such as a refund window or a delivery guarantee, also requires sign-off from the Head of Customer Experience.

When this handbook conflicts with a signed customer contract, the contract wins. When it conflicts with local law, the law wins, and the conflict must be reported to the Legal team within five business days so the handbook can be corrected.

2. Company Overview

Northwind Freight was founded in 2011 in Fort Worth, Texas, as a regional pallet carrier serving four counties. It now operates nine distribution centers across Texas, Oklahoma, and Louisiana, and moves roughly 14,000 shipments per day during peak season. The largest facility is the Alliance hub, which handles about 38 percent of total volume.

The company has three customer segments. Retail partners ship store replenishment orders on fixed weekly schedules. E-commerce merchants

ship parcels directly to consumers and care most about delivery speed and tracking accuracy. Industrial accounts ship heavy or oversized freight and usually negotiate custom rates.

Operations are organized into four groups: Linehaul, which moves trailers between hubs; Last Mile, which handles final delivery; Warehouse, which receives, stores, and picks freight; and Customer Experience, which owns support, claims, and refunds. Engineering supports all four groups and owns the dispatch platform, the tracking API, and the customer portal.

3. Refund and Claims Policy

Customers may request a full refund for any shipment that was not delivered within the guaranteed window for its service tier. Refund requests must be submitted within 30 days of the original delivery date. Requests received after 30 days are reviewed case by case and are usually converted to account credit rather than a cash refund.

Damage claims follow a different process. The customer must report visible damage within 48 hours of delivery and provide at least two photographs of the damaged goods and packaging. Concealed damage, meaning damage that was not visible until the package was opened, may be reported within 7 days. Claims without photographs are not rejected automatically, but they are routed to a senior claims specialist for manual review.

Support agents may approve refunds up to \$500 on their own authority. Refunds between \$500 and \$5,000 require approval from a support team lead. Any refund above \$5,000, and any exception to the 30-day window, must be approved by the regional manager. Every approval is recorded in the claims system with the approver's employee ID so that audits can trace each decision.

Refunds are issued to the original payment method within 10 business days of approval. Account credits are applied immediately and expire after 12 months. Agents must never promise a refund amount to a customer before the approval step is complete, because partial refunds are common for shipments that were delayed but still delivered.

4. Service Tiers and Delivery Guarantees

Northwind offers four service tiers. The table below summarizes the guaranteed delivery windows and the base rates for a standard parcel under 20 pounds. Rates for heavier parcels and pallets are calculated by the pricing engine and are not listed here.

Tier	Guaranteed window	Base rate	Tracking
Economy	5 business days	\$8.50	Daily updates
Standard	3 business days	\$12.00	Real-time
Express	Next business day	\$24.00	Real-time
Priority	Same day (DFW only)	\$49.00	Real-time + SMS

The guarantee applies only to shipments picked up before the daily cutoff of 4:00 PM local time. Shipments picked up after the cutoff start their guarantee window the next business day. Weather events declared by the National Weather Service, and road closures ordered by a government agency, suspend all guarantees for the affected region until the event is lifted.

Saturday delivery is available for Express and Priority shipments in metro areas for an additional \$9.00. Sunday delivery is not offered. Same-day service is limited to shipments that both originate and terminate inside the Dallas-Fort Worth metro area.

5. Warehouse Safety

Safety rules apply to everyone on the warehouse floor, including visitors and engineers on site for equipment installs. High-visibility vests and steel-toe boots are mandatory in all marked zones. Hearing protection is required in the sortation area whenever the conveyor system is running, because noise levels there regularly exceed 85 decibels.

Only certified operators may drive forklifts or reach trucks. Certification is valid for three years and requires a written test and a supervised driving assessment. An operator who is involved in any collision, including a minor one with no injury, must stop work, report the incident to the shift supervisor, and complete a refresher assessment before operating equipment again.

Pallets may not be stacked higher than 12 feet in the bulk storage area, and racking must never be loaded beyond the weight rating printed on the beam label. Damaged racking is tagged red and unloaded within one shift. Fire exits and the 18-inch clearance below sprinkler heads must be kept clear at all times; blocked exits are treated as a serious safety violation.

6. Data Handling and Retention

Northwind stores personal data about shippers and recipients, including names, addresses, phone numbers, and delivery photos. This data may only be accessed for a legitimate operational purpose such as completing a delivery, resolving a claim, or responding to a support request. Browsing customer records out of curiosity is a disciplinary offense.

Delivery photos are retained for 90 days and then deleted automatically. Shipment records, including addresses and tracking events, are retained for 7 years to meet tax and audit requirements. Support call recordings are kept for 180 days. Application logs must never contain full addresses, phone numbers, or payment details; engineers are expected to mask these fields

before logging.

Customers may request a copy of their personal data or ask for it to be deleted. These requests are handled by the Privacy team and must be acknowledged within 3 business days and completed within 30 days. Deletion does not apply to shipment records under legal retention, but those records are restricted so that only the Finance and Legal teams can access them.

Access to production databases requires a ticket that names the specific dataset and the business reason. Access is granted for a maximum of 8 hours and is revoked automatically. Bulk exports of customer data require written approval from the Data Protection Officer.

7. Dispatch Platform Overview

The dispatch platform assigns shipments to routes and drivers. Each night at 11:00 PM it runs a batch optimization that builds the next day's routes, balancing driver hours, vehicle capacity, and delivery windows. During the day, new same-day and Express shipments are inserted into existing routes by an incremental planner that runs every 15 minutes.

The platform exposes a tracking API used by the customer portal and by large e-commerce merchants. The API is rate limited to 600 requests per minute per API key. Merchants who need a higher limit must request it through their account manager, and the increase is approved by the Platform Engineering lead after a load review.

Route plans are stored in PostgreSQL and cached in Redis for fast lookups by driver devices. If the cache is unavailable, driver devices fall back to reading directly from a read replica, which is slower but keeps deliveries moving. A full platform outage triggers the manual dispatch procedure described in section 9.

8. On-Call Responsibilities

Engineering runs a weekly on-call rotation that starts every Monday at 10:00 AM Central Time. Each rotation has a primary and a secondary engineer. The primary must acknowledge a page within 5 minutes during business hours and within 15 minutes outside business hours. If the primary does not acknowledge in time, the page escalates to the secondary, and after a further 15 minutes it escalates to the engineering manager.

On-call engineers receive a stipend of \$300 per week and are not expected to do planned project work during their rotation. Instead, they work through the operational backlog: flaky alerts, runbook gaps, and small reliability fixes. At the end of the week, the outgoing primary writes a handoff note listing open issues, recent changes, and anything unusual the next person should watch.

An engineer may not be on call for more than one week in any four-week period, except by volunteer swap. Swaps must be recorded in the on-call scheduling tool at least 24 hours before the rotation starts so that the paging system routes alerts to the right person.

9. Incident Response

Incidents are classified by severity. A SEV1 is a full outage of the dispatch platform or the tracking API, or any event that stops deliveries in more than one region. A SEV2 is a partial outage or a serious degradation affecting a single region or a single major customer. A SEV3 is a minor issue with a workaround. Data exposure of any size is always treated as a SEV1, regardless of how many records are involved.

For a SEV1, the on-call engineer declares the incident in the incident channel, becomes the incident commander until relieved, and pages the communications lead. The communications lead posts the first

customer-facing status update within 30 minutes and updates it at least every hour until the incident is resolved. The incident commander's job is coordination, not debugging; they assign investigators and keep a written timeline.

If the dispatch platform is down for more than 45 minutes during operating hours, hub managers switch to manual dispatch. Each hub keeps printed route sheets generated by the previous night's batch run, and drivers follow those sheets while the platform is restored. Same-day insertions are paused during manual dispatch.

Every SEV1 and SEV2 requires a written postmortem within 5 business days. Postmortems are blameless: they describe what happened, why the system allowed it to happen, and what will change. Each postmortem must produce at least one action item with a named owner and a due date, and open action items are reviewed at the weekly operations meeting.

10. Onboarding Checklist

New hires in operations roles complete a two-week onboarding program. In the first week they shadow an experienced team member, complete the safety course, and read this handbook. In the second week they handle real work under supervision, starting with low-risk tasks such as tracking inquiries before moving to claims and refunds.

Engineers follow a separate onboarding track. By the end of their first week they should be able to run the dispatch platform locally, deploy a change to the staging environment, and explain the on-call escalation path. They join the on-call rotation as a shadow after six weeks and as a secondary after ten weeks. Nobody is placed as primary on-call before completing three shadow shifts.

Every new hire is assigned an onboarding buddy who is not their manager. The buddy meets with them at least twice a week during the first month and is the first point of contact for questions that feel too small to ask a manager.

11. Sustainability Commitments

Northwind has committed to reducing emissions per shipment by 40 percent from its 2022 baseline by 2030. The main levers are electric delivery vans in dense urban routes, better route optimization to reduce empty miles, and solar panels on the roofs of the three largest distribution centers.

As of the last annual report, 22 percent of the last-mile fleet was electric. Electric vans are prioritized for routes under 90 miles per day, because that keeps them comfortably within battery range without midday charging. Diesel tractors remain in use for linehaul, where electric alternatives are not yet practical for the distances involved.

Packaging waste is handled at the hub level. Each distribution center recycles cardboard and shrink wrap, and damaged pallets are sent to a repair partner rather than discarded. Customers can request carbon reports for their shipments through the customer portal.

12. Glossary

Cutoff: the latest pickup time, 4:00 PM local, for a shipment to start its guarantee window the same day. Linehaul: the movement of full trailers between hubs, usually overnight. Last mile: the final leg from a local hub to the recipient's door.

Manual dispatch: the fallback procedure in which drivers follow printed route sheets when the dispatch platform is unavailable. Incident commander: the person who coordinates an active incident; this is a coordination role and does not require being the most senior engineer present.

Postmortem: a blameless written review of an incident, due within 5 business days for SEV1 and SEV2 events. Concealed damage: damage that is not visible until a package is opened, reportable within 7 days of delivery.